

Nayan Bhatia

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github.com/nayanbhatia311 | [Google Scholar](https://scholar.google.com/citations?user=...)

Summary

PhD candidate in Computer Science and Engineering building machine learning systems that move from research prototypes to deployed products. Recent work includes transformer models for wireless sensing, LLM evaluation and infrastructure at Zscaler, and a UC-wide AI application serving roughly 300,000 students. Expected graduation in March or June 2027.

Technologies

Programming: Python, C/C++, SQL, Java, Swift, shell scripting

Machine Learning: Transformers, LLMs, PyTorch, PyTorch Lightning, TensorFlow, scikit-learn, transfer learning

AI Application Stack: Data ingestion/processing, training and inference pipelines, APIs, model serving, user-facing interfaces

Data & Evaluation: NumPy, Pandas, Pydantic, dataset curation/validation, benchmark design, regression testing

Systems: Distributed systems, Linux, Docker, Kubernetes, Slurm, AWS, GCP, Git/GitHub Actions, CI/CD

Education

University of California, Santa Cruz Sep 2021 – Mar/Jun 2027
 PhD Candidate in Computer Science & Engineering (transitioned from M.S) (flexible)

[Inter-Networking Research Group \(i-NRG Lab\)](#) under Prof. Katia Obraczka

K.J. Somaiya Institute of Engineering and IT, University of Mumbai Aug 2017 – Jun 2021
 B.E. in Computer Engineering

Experience

Zscaler, San Jose, CA Summer 2026
ML/AI Intern – Manager: Dat Nguyen

- Built an automated Python/Pydantic evaluation framework for LLM systems on Terminal-Bench 2.1, comparing routing and context-compression methods with repeatable quality and cost metrics; the selected optimization was projected to reduce costs by 20%.
- Deployed production LLM services for Zscaler's internal LiteLLM gateway using Kubernetes, Helm, Argo CD, Prisma Migrate, and AWS ECR/ECS/EKS.
- Established recurring evaluation and regression testing for the selected optimization, monitored behavioral changes, and maintained reproducible containerized releases across environments.

University of California, Santa Cruz Oct 2021 – Present
LLM & Platform Engineer

- Developed the **UC Basic Needs platform** (ucbasicneeds.ucsc.edu) and **UC Essential Needs Navigator**, a UC-wide AI application serving ~300,000 students across 10 campuses; [featured by UCSC News](#).
- Led the end-to-end stack as primary developer: structured-data ingestion and processing, APIs, an LLM-powered natural-language interface with cited results, cloud deployment, and stakeholder collaboration.

University of California, Santa Cruz 2021 – Present
Teaching Assistant

- Taught Python, C, IoT, database systems, and operating systems; explained systems concepts and debugging workflows to cohorts of 100+ students per quarter.

Research & Projects

Argus: Scalable Wi-Fi Person Identification 2025 – 2026

- Developed an attention-guided decoder-only Transformer and reproducible data, training, and evaluation pipelines for a 154-subject dataset, using leakage-resistant splits and segment-level aggregation.
- Reached 84.85% Top-1 and 99.26% Top-5 accuracy on 60-second segments; improved over a raw-CSI Transformer by 7.75 points with $4.4\times$ fewer inference FLOPs per window.

Locaris: LLM-based Indoor Localization from Wi-Fi Telemetry 2025 – Present

- Developed a compact decoder-only Transformer that learns directly from raw heterogeneous telemetry (CSI, RSSI, and FTM) without handcrafted feature pipelines and predicts device position.
- Achieved sub-meter RSSI/FTM localization and centimeter-level CSI results across controlled testbeds; designed transfer-learning workflows for unseen devices and deployments.

Network Simulation Bridge (NSB) 2023 – Present

- Co-developed an open-source framework connecting distributed applications to OMNeT++ and ns-3 through reusable interfaces; published at ACM Q2SWinet and presented as an IEEE ICDCS tutorial.

PulseFi: Contactless Vital-Signs via Commodity Wi-Fi 2024 – Present

- Built an end-to-end sensing and ML pipeline on ESP32/Raspberry Pi devices, validating live heart- and breathing-rate inference against medical-grade reference sensors.

Optimystic MVA: Visual Acuity iOS App 2026

- Developed an open-source Swift iOS app for measuring visual acuity and published the [source code](#) for users to compile and run independently.

Selected Publications

- N. S. Bhatia et al. *ARGUS: Attention-Guided Transformers for Scalable Person Identification Using Wi-Fi Telemetry*. arXiv, 2026. [Paper](#).
- N. S. Bhatia et al. *Indoor Localization Using Compact, Telemetry-Agnostic, Transfer-Learning Enabled Decoder-Only Transformer*. IEEE PerCom 2026 WiP. [arXiv](#) | [UC patent disclosure](#).
- P. Kocheta, N. S. Bhatia, and K. Obraczka. *Pulse-Fi: A Low-Cost System for Accurate Heart Rate Monitoring Using Wi-Fi Channel State Information*. IEEE DCOSS-IoT 2025. [PDF](#).

Awards & Honors

- **QB3 Santa Cruz Graduate Innovators Fellowship (2026)**: Awarded for advancing PulseFi, a WiFi-based vital signs monitoring system with commercialization potential in healthcare.
- **Technical Program Committee Member**, IEEE LCN 2026.
- **Peer Reviewer**: IEEE Transactions on Mobile Computing (2025, 2026) and ACM TOMPECS (2026).
- **Young Gladiators, Colosseum Master Class (DARPA/NSF)**: Invited for training on large-scale wireless experimentation using the Colosseum emulator.